Math 241, F12 **Name:** Answer

Quiz 7 **SSN:** xxx-xx-

<u>Instructions:</u> There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (6 points)

(a) Find the first partial derivatives of the function $f(x, y) = e^{-x^2y}$.

$f_x(x,y) = -2xye^{-x^2y}$

$f_y(x,y) = -x^2e^{-x^2y}$

(b) Find the second partial derivatives of the function $z = x^3y^2 + 2x^4y$.

$\frac{\partial z}{\partial x} = 3x^2y^2 + 8x^3y \qquad \frac{\partial z}{\partial y} = 2x^3y + 2x^4$

$\frac{\partial^2 z}{\partial x^2} = 6xy^2 + 24x^2y \qquad \frac{\partial^2 z}{\partial y \partial x} = 6x^2y + 8x^3$

$\frac{\partial^2 z}{\partial x \partial y} = 6x^2y + 8x^3 \qquad \frac{\partial^2 z}{\partial y^2} = 2x^3$

Problem 2. (4 points) Find an equation of the tangent plane to the surface $z = xy$ at the point $(1, 1, 1)$.

Let $f(x,y) = xy$

$f_x(x,y) = y, \quad f_y(x,y) = x$

$\Rightarrow f_x(1,1) = 1, \quad f_y(1,1) = 1$

Thus an equation of the tangent plane at $(1,1,1)$ is

$z - 1 = 1\cdot(x-1) + 1\cdot(y-1)$

$z = x + y - 1.$